HMGP 7622

Rethinking data analysis

A researcher's guide to avoiding missteps and misuse

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Day 03

Error types and Statistical power

- Type I and Type II errors in practice
- Statistical power: the underpowered majority
- Sample size planning: beyond ritual

Statistical power of a study

The statistical power of a study is the probability that it can distinguish an effect of a given size from random chance.

Power = True positive rate = Sensitivity = Recall

Many studies are underpowered → Waste of resources & Unethical

http://www.nature.com/doifinder/10.1038/nmeth.2738

Type I & type II error

Type I error: False-positive rate (α)	**Type II error:** False-negative rate (β)
Rejecting the null hypothesis when it is actually true "Seizing on random signal"	Failing to reject the null hypothesis when an alternative hypothesis is true "Missed opportunity"

Remember the story of the boy that cried wolf! The villager make Type I error at the beginning & Type II error at the end!

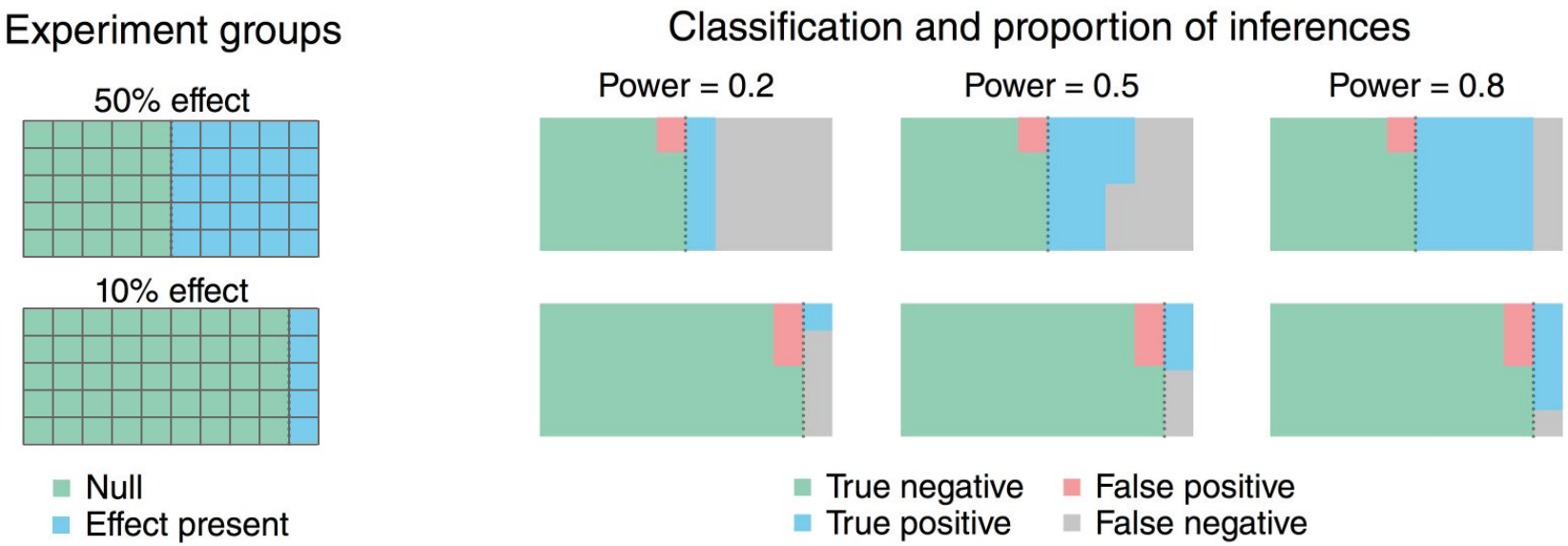
Statistical power

Definition

Probability that the study can distinguish an effect of a given size from random chance.

Probability of correctly rejecting the null hypothesis when a specific alternative is true.

Power = True positive rate = Sensitivity = Recall



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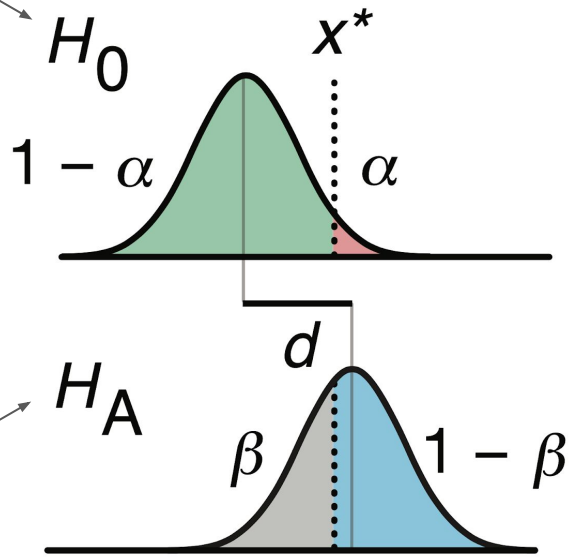
Statistical power

Definition

True negative
False positive
True positive
False negative

Null hypothesis



Alternative hypothesis

Correct inference
- Specificity, $1 - \alpha$
- Power, sensitivity, $1 - \beta$

Incorrect inference
- Type I error, α
- Type II error, β

Krzywinski & Altman (2013) Nat. Meth.

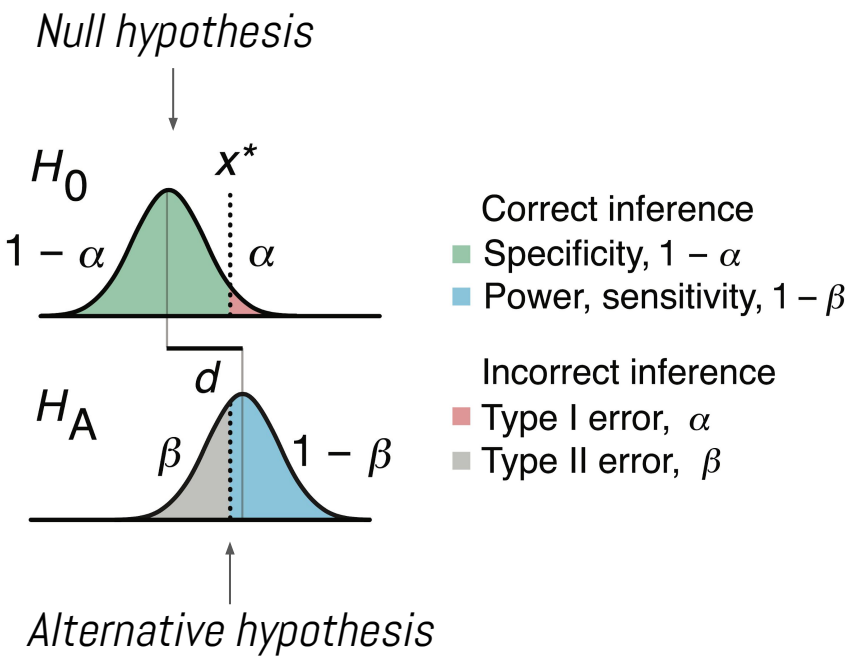
Statistical power

Definition

The power is the probability that the test correctly rejects the null hypothesis (H_0) when a specific alternative hypothesis (H_1) is true.

- Pr(reject H_0 | H_1 is true)
- H_1 has to be specific (cannot just be negation of H_0)

As power ($1 - \beta$) increases ⟶ Probability of making type II error (β) decreases.



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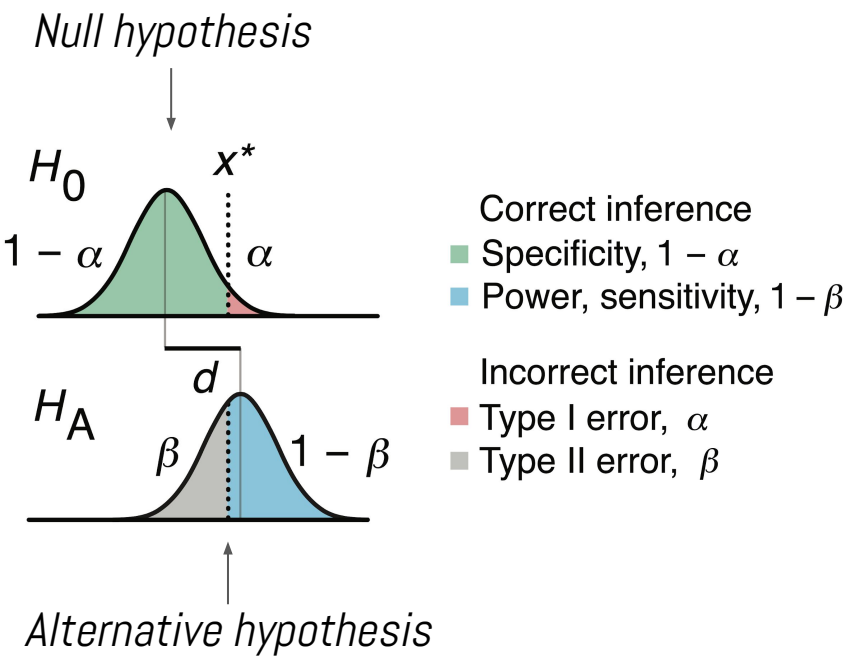
Statistical power

Trade-off, thresholds, & asymmetry of fear

- Good to have low α (FPR) & low β (FNR):
  - But, the α and β rates are inversely related: ↓α → ↑β (& reduces power).
- Typically, α < β
  - Consequences of FP (in an extreme case, a retracted paper) are more serious than those of FN (a missed opportunity to publish).
- But, the balance between α and β depends on the objectives:
  - If FP are subject to another round of testing but FN are discarded, β should be kept low.



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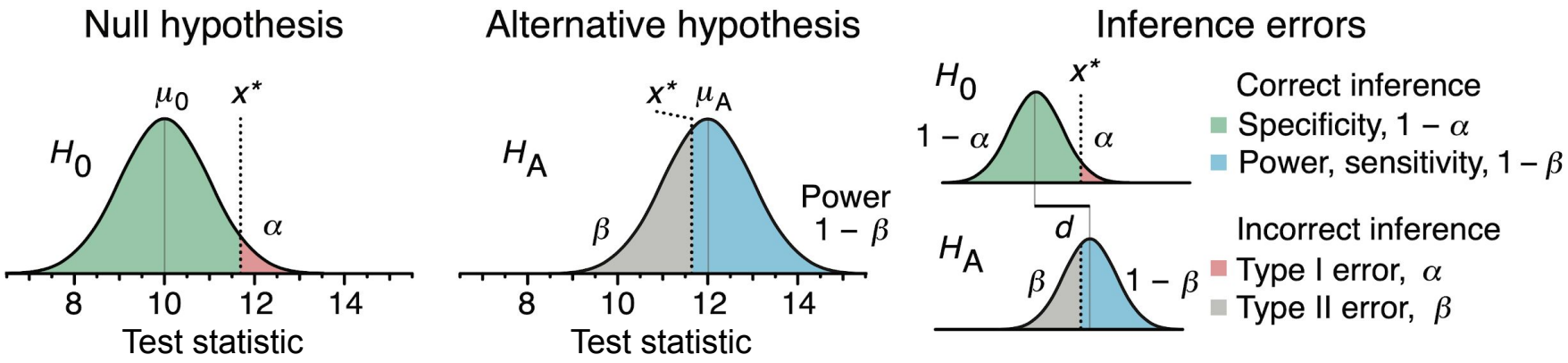
Statistical power

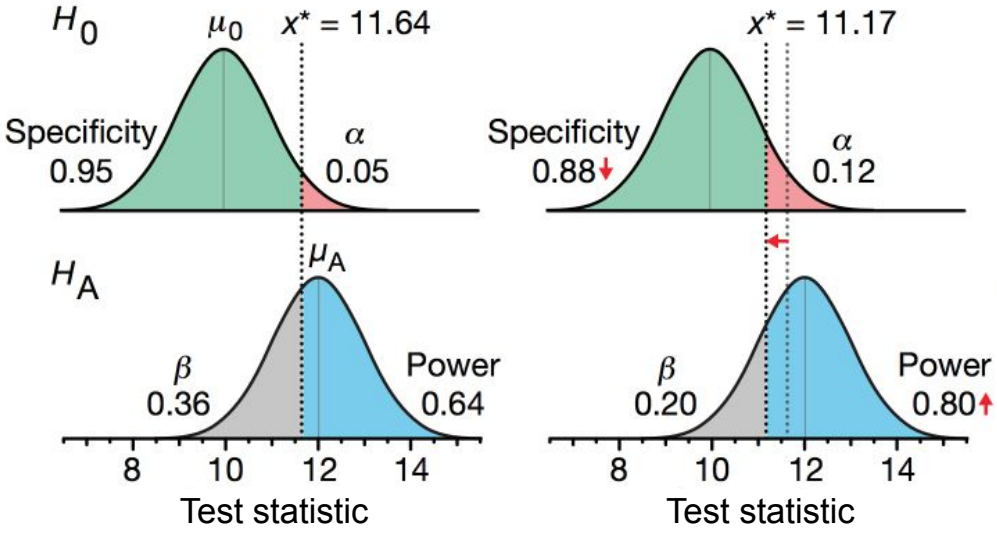


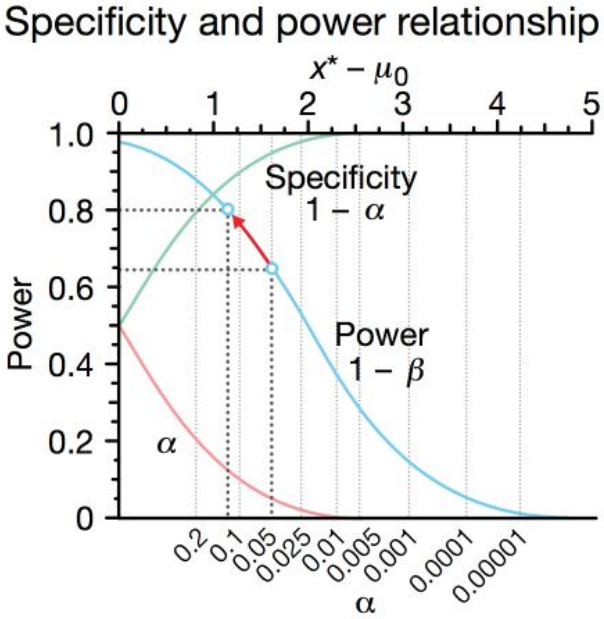
- Values sampled from $H_A < x^*$ do not trigger rejection of H_0 and occur at a rate β.
- Power (sensitivity; TPR) = 1 – β (blue area).
- Good to have low α (FPR) & low β (FNR), but:
  - The α and β rates are inversely related: ↓α → ↑β (& reduces power).

Krzywinski & Altman (2013) Nat. Meth.

Compromise between specificity and power

Decreasing specificity (TNR) increases power (TPR)





Can we improve our chance to detect increased effect from H_A (increase power) without compromising **α** (increasing FP)?

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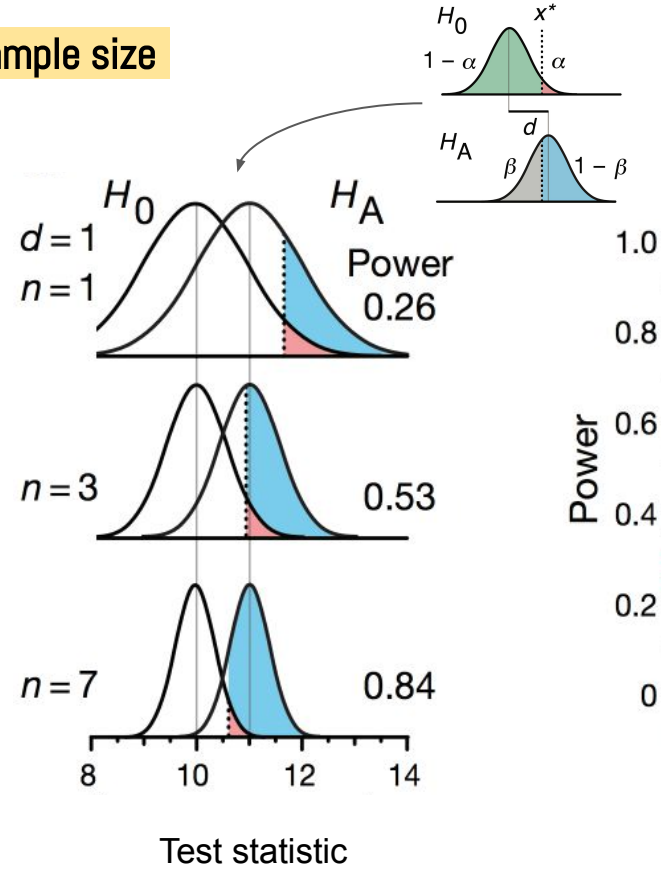
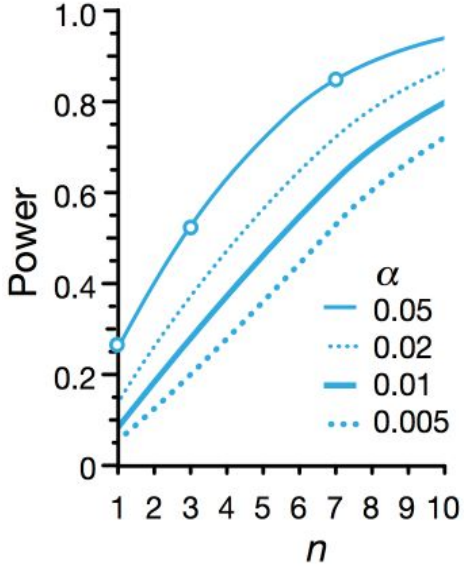
Factors affecting power

Sample size





Control experimental conditions (e.g. using genetically identical orgs under lab conditions; adding precise amount of a drug) to reduce the variation b/w samples & compensate to some extent for small sample sizes.

In practice, because we estimate population σ from the samples, power is decreased and we need a slightly larger sample size to achieve the desired power.

Krzywinski & Altman (2013) Nat. Meth.

Factors affecting power

Power depends on multiple factors

- Statistical significance criterion:
  - Lesser conservative test (larger significance criterion) ⟶ More power
- Sample size:
  - Collecting more data ⟶ Easier to detect small effects; relates to the efficiency of a given testing procedure, experimental design, or an estimator (sample size required for a given power)
- Size of the effect:
  - Larger the effect ⟶ Easier it is to detect; (std. effect size better)

- Measurement error: counting cells vs. estimating level of fatigue/depression
- Experimental design: e.g. in a two-sample setting, optimal to have equal number

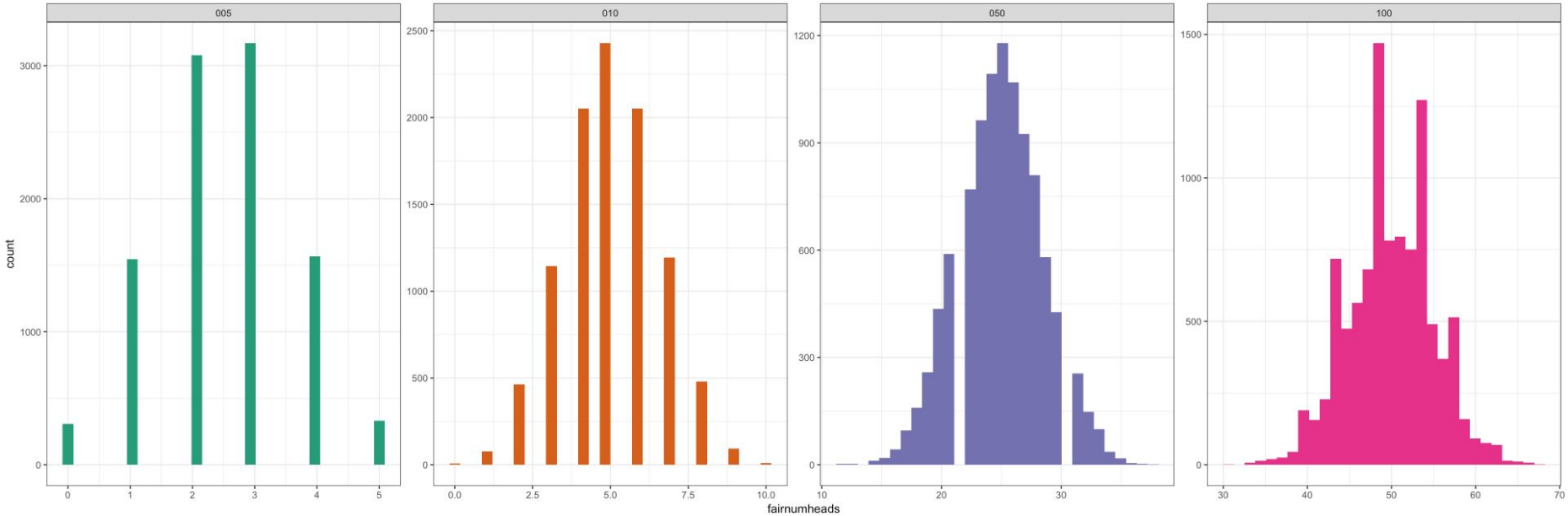
Factors affecting power

Generating a power curve

Let's examine some code to generate a power curve to detect unfair coins:

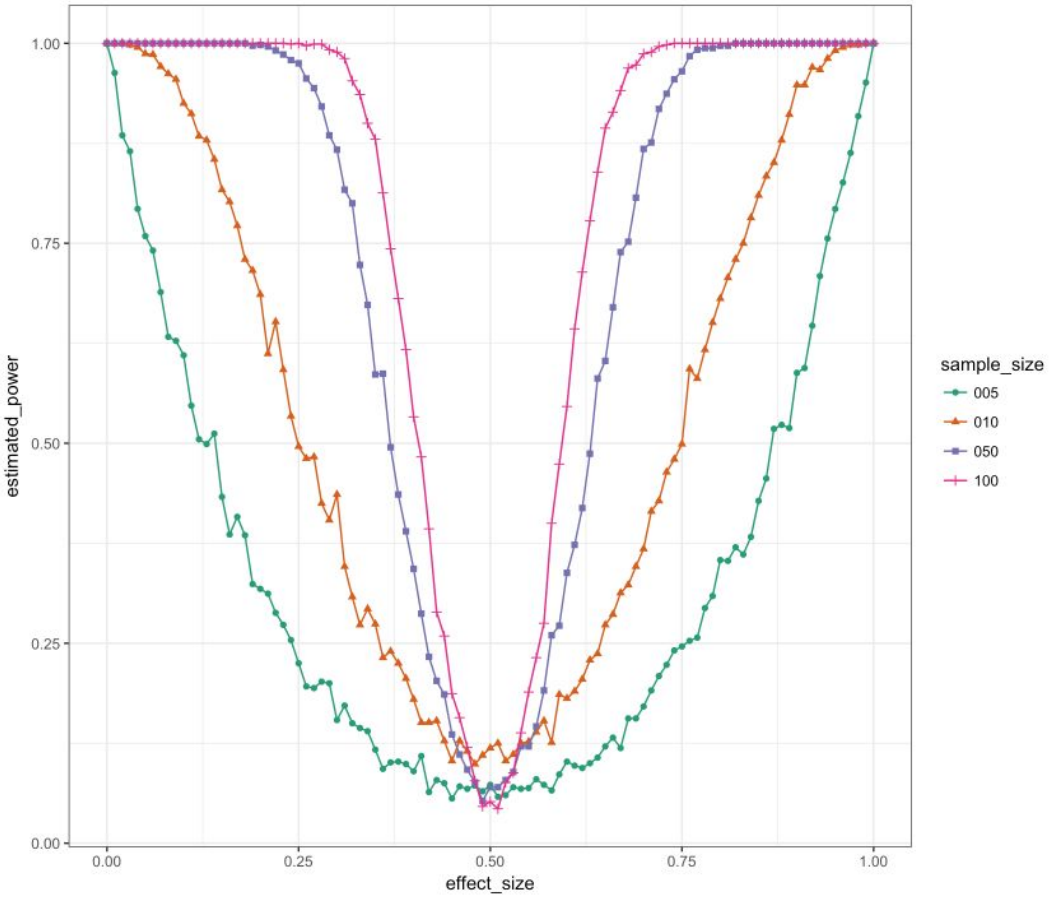
- You are given a coin and asked to detect if it is biased.
- Experiment: Flip the coin **num_flips** and take a call.
- Establish the null hypothesis (**num_permutations** = 10,000) for a given **num_flips** (this is the **sample size**).
- For a given **bias** (i.e. the **effect size**), find out how many times does an experiment like the one above can reject the null hypothesis.

Factors affecting power



Null distributions for different sample sizes

Factors affecting power



Power analysis

Balancing sample size, effect size, and power

Critical to good study design

- First, set the values of type I error (α) and power ($1 - \beta$) to be statistically adequate:
  - Traditionally 0.05 and 0.80, respectively.
- Then determine sample size (n) on the basis of the smallest effect we wish to measure.
  - If the required sample size is too large → may need to reassess objectives or more tightly control the experimental conditions to reduce the variance.
- When the power is low, only large effects can be detected, and negative results cannot be reliably interpreted.

Underpowered studies

Lower chance of detecting a true effect

Lower power ⟶ lower recall (true positive rate).

Low-powered studies produce more FN than high-powered studies.

- When studies in a given field are designed with a power of 20%, it means that if there are 100 genuine non-null (i.e. real) effects to be discovered in that field, these studies are expected to discover only 20 of them.

Button et al., (2013) Nat Rev Neuro

Underpowered studies

Lower likelihood that a statistically significant result reflects a true effect

Lower power ⟶ lower precision (positive predictive value; PPV) when an effect is claimed.

Button et al., (2013) Nat Rev Neuro

Underpowered studies

Lead to exaggerated estimate of the magnitude of the effect when true effect is discovered

Example scenario

An association *truly exists* with an **effect size of ~1.20** .

We're trying to discover it by performing a small study with power of, say, ~20%.

The results of any study are subject to sampling variation and random error in the measurements of the variables and outcomes of interest.

- The study may find an effect <1.20 (e.g., **1.00**) or >1.20 (e.g., **1.60**).
- Effect of 1.00 or 1.20 will not reach statistical significance because of the small sample size.
- Nominally significant association only when random error creates an effect of 1.60.

Button et al., (2013) Nat Rev Neuro

Underpowered studies

Winner's curse

The 'lucky' scientist who makes the discovery in a small study is cursed by finding an inflated effect.

- Likely to occur whenever claims of discovery are based on thresholds of statistical significance (for example, $p < 0.05$) or other selection filters.

Worst for small, low-powered studies, which can only detect effects that happen to be large.

- E.g. if the true effect is medium-sized, only those small studies that, by chance, overestimate the magnitude of the effect will pass the threshold for discovery.

Example scenario

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Button et al., (2013) Nat Rev Neuro

Underpowered studies

Hamper replicating research findings

If the original estimate of the effect is inflated, then replication studies will tend to show smaller effect sizes as findings *converge* on the true effect.

More replication studies → eventually arrive at the more accurate effect size, but this may take time or may never happen if we only perform small studies.

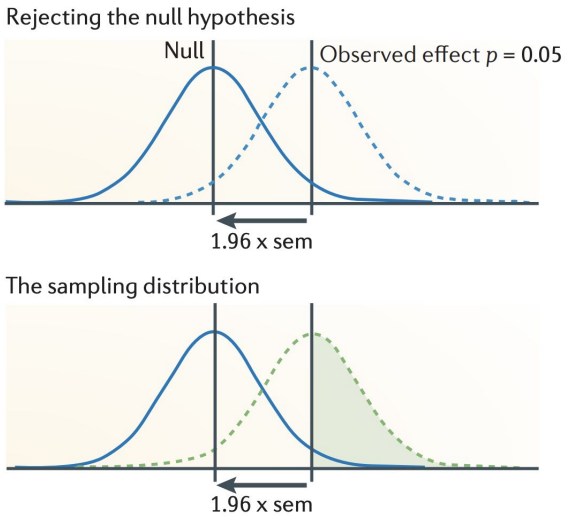
Button et al., (2013) Nat Rev Neuro

Underpowered studies

Hamper replicating research findings

Common misconception: A replication study will have sufficient power to replicate an initial finding if the sample size is similar to that in the original study.

A replication study that uses the same sample size as the original study will only achieve ~50% power, even if the original study accurately estimated the true effect size.



- If sample sizes are determined by historical precedent (rather than through formal power calculation) → place an upper limit on average power within that field.
- Sample size estimate using even using power calculation will be overly optimistic if based on estimates of the 'true effect size' (inflated owing to winner's curse).

Button et al., (2013) Nat Rev Neuro

Underpowered studies

Underpowered studies are wasteful & unethical

Even a study that achieves only 80% power still presents a 20% possibility that the animals have been sacrificed without the study detecting the underlying true effect.

- If the average power of studies is 20–30%, the ethical implications are substantial.

Continuing data collection once it is clear that the effect being sought does not exist or is too small to be of interest is probably unethical.

  - Studies are not just wasteful when they stop too early, they are also wasteful when they stop too late.

Button et al., (2013) Nat Rev Neuro

Sample size planning: beyond ritual

Perform an a priori power calculation

- Use existing literature to estimate the size of effect and potential study design. But exercise caution:
  - Common mistake: Designing studies based on inflated effect sizes found in existing literature.
    - Because they're usually over-optimistic/inflated effects
- Perform a priori power analysis
  - Determine sample sizes *before* data collection based on the smallest effect of practical interest, the desired power, and the significance level.
  - Post-hoc power calculations are essentially useless. Using the estimates provided by the just-collected data to calculate power invariably inflates the result and provides a false sense of security.
- When doing the analysis:
  - Don't ignore attrition: Failing to account for dropouts, non-responders, or missing data reduces the final effective sample size.
  - Adjust for the inflation of Type I errors when testing many outcomes.
- If time or financial constraints mean your study is underpowered, make this clear and acknowledge this limitation (or limitations) in the interpretation of your results.

Button et al., (2013) Nat Rev Neuro

Sample size planning: beyond ritual

Disclose methods and findings transparently

- If the intended analyses produce null findings and you move on to explore your data in other ways, say so.
- Null findings locked in file drawers bias the literature, whereas exploratory analyses are only useful and valid if you acknowledge the caveats and limitations.

Button et al., (2013) Nat Rev Neuro

Sample size planning: beyond ritual

Pre-register your study protocol and analysis plan

- Pre-registration clarifies whether analyses are confirmatory or exploratory, encourages well-powered studies and reduces opportunities for non-transparent data mining and selective reporting.

Work collaboratively to increase power and replicate findings

- Combining data increases the total sample size (and therefore power) while minimizing the labour and resource impact on any one contributor.
- Large-scale collaborative consortia in fields such as human genetic epidemiology have transformed the reliability of findings in these fields.

Button et al., (2013) Nat Rev Neuro

Typical sample sizes

- Clinical research (behavioral or drug treatments):
  - Need enough participants to represent all subtypes for which treatment might be used.
  - Some issues: lack reliable methods for diagnosis.
  - Rough rule of thumb: at least 100 people.
    - The actual number needed to find a valid effect depends on a range of factors, including the magnitude and frequency of the effect in the general population.

https://www.spectrumnews.org/news/studies-size-matters-let-us-count-ways/

Typical sample sizes

- Brain imaging studies:
  - Historically included 20 or fewer participants. In the past 10 years, closer to 100 participants.
  - Studies that aim to trace developmental trajectories should also track the same few individuals over time, scanning their brains at regular intervals, rather than examining a cross-section of people of different ages at different sites.

https://www.spectrumnews.org/news/studies-size-matters-let-us-count-ways/

Typical sample sizes

- Genetic studies (large no. of variants/genes, each making a small contribution):
  - Rare variants in coding regions: order of thousands of people.
  - Risk variants across the whole-genome: tens of thousands of individuals.
    - Millions of statistical tests, one per variant → increases FPR.
  - GWAS: hundreds of thousands of individuals
    - Common gene variants that contribute to the risk of a condition.

https://www.spectrumnews.org/news/studies-size-matters-let-us-count-ways/

Typical sample sizes

- Preclinical research:
  - Underpowered animal studies for decades (cost and ethical issues).
  - Make up for their low numbers by analyzing a large number of cells or other samples from each animal → 'pseudoreplication.'
  - Can control lab animals' diets, ages and housing conditions, and scale doses or treatments by weight → sample sizes on the order of 10 animals to be acceptable. Should ≥15 per group to identify important biological effects.
  - In the past few years, push for larger numbers in animal studies.

https://www.spectrumnews.org/news/studies-size-matters-let-us-count-ways/

Typical sample sizes

- Biomarker studies (physiological characteristics, such as patterns of eye movements, brain waves or activity, or blood chemistry):
  - Candidate biomarkers have often failed in subsequent studies.
  - Must draw samples from at least 100 individuals.
  - Clinical trials of biomarkers designed to flag people with disease → ≥ 1,000 participants. Researchers should also replicate the efficacy of a biomarker in an independent sample.
  - Some scientists are designing biomarker studies of thousands of participants that combine data from behavioral, imaging and genetic studies.

https://www.spectrumnews.org/news/studies-size-matters-let-us-count-ways/

Typical sample sizes

- Field trials:
  - Variables that are hard to control, and so must include hundreds of individuals to yield meaningful results.
  - Needs more than an appropriate number of participants.
    - Representative mix of sexes and ages.

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